

Week 3: Algebraic Knowledge

Jacob Khohayting

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※ Setting Variables

A squirrel has a collection of acorns. It finds 12 more acorns, and now it has 27. How many acorns did the squirrel start with?

This problem could be interpreted as a subtraction problem. To get the start, we subtract 12 from 27 to get 15. However, it can also be interpreted as an addition problem.

In many math problems, there is a “missing” piece of information. Instead of guessing or working backward, we can give that missing piece a name - like x or a symbol like $\square$ or $\triangle$ - and treat it like a real number. This is called setting a variable.

Here, we can set a variable for the number of acorns the squirrel started with as A . Then, $A + 12 = 27$. So, subtracting 12 yields $A = 15$.

For this problem, it might feel useless. But, it is a good starter to get a hang of how setting variables should work. They will usually help you get a grasp of a problem with far more ease. On many occasions, it is hard to determine what to set a variable for. This is what this lesson is generally for. Here are some starting tips:

- Define your variable clearly: Always write down what your letter stands for (e.g., "Let x = the number of apples").
- Translate word-for-word: “Sum” means add, “Product” means multiply, “Difference” means subtract, “Quotient” means divide.
- The Golden Rule of Equations: Whatever you do to one side of the equals sign, you must do to the other side to keep it balanced.
- Check your work: Take your answer and plug it back into the original word problem. Does it make sense?

※ Quick Practice

I am thinking of a number. If I multiply it by 4 and then subtract 7, I get 25. What is my number?

Setting variables are good for getting a grasp. Looking at the problem, you might start to guess-and-check. Surely, you would get there. But, that might take a while.

Instead, we want to set a variable. Variables are good to set for things you don’t know or want to figure out. Here, we want to find the number. So, let the variable for the number be n . Then, multiplying by 4 gives $4n$ and then subtracting 7 gives $4n - 7$. Equating means $4n - 7 = 25$.

Now, the algebra is much easier. Solving this, you can add 7 to both sides and then divide by 4 to get 8!

Another way to see this problem without variables is to go backwards. If you end at 25. Then, by subtracting 7 you must’ve started with 32. Multiplying by 4 to get 32 means you started with 8!

In a jar of marbles, the ratio of Red marbles to Blue marbles is 3 : 5. If there are 40 marbles in total, how many are Red?

Here, we want to find the number of red marbles. We can set a variable for that. Suppose there are r red marbles. Then, the number of blue marbles is $\frac{5}{3}r$. The total marbles is

$$r + \frac{5}{3}r = \frac{8}{3}r = 40 \implies r = 15.$$

You can also set a variable for the blue marbles. But, for now, we are just dealing with one-variable equations.

Sarah is 3 years older than her brother Tom. In 5 years, the sum of their ages will be 31. How old is Tom now?

Let T be Tom's age now. Notice, Sarah is 3 years older so Sarah's age is $T + 3$. In five years, Tom's and Sarah's ages would be increased by 5 so $T + 5$ and $T + 8$. The sum is:

$$(T + 5) + (T + 8) = 2T + 13 = 31 \implies T = 9.$$

※ Substitution Method

For these problems, you may find some sort of system of equation such as the following:

$$A + B = 15, \text{ and}$$

$$A - B = 5.$$

There are two main types of ways to solve this problem. We will focus on one here and another way in the next section. For this, we want to isolate a variable at all times. For instance, $A - B = 5$ can be converted to $A = B + 5$ by adding B on both sides. Now, we substitute $A = B + 5$ into the first equation to get:

$$A + B = (B + 5) + B = 15 \implies 2B + 5 = 15.$$

Solving for B , we get $B = 5$. Tracing back our steps then $A = 10$.

In general, we always want to isolate a variable. After we isolate, we substitute to eliminate that variable. Each time, the number of variables we are dealing with decreases. Once we have the value of a variable, we can trace back.

In the above example, we substituted for A to eliminate A . Then, we had a one-variable equation involving B which isn't too hard to solve.

This might be confusing, but let's try an example with 3 variables:

$$x + y + z = 4,$$

$$x - y + z = 2, \text{ and}$$

$$x + 3y + 4z = 9.$$

Let's outline this solution step by step to get a better idea:

- **Isolate x :** Set $x = y - z + 2$ with the second equation.
- **Substitute x into the other equations:** Substituting into the first equation,

$$(y - z + 2) + y + z = 4 \implies 2y + 2 = 4 \implies y = 1.$$

For the other equation,

$$(y - z + 2) + (3y + 4z) = 9 \implies 4y + 3z = 7.$$

Fortunately, we actually obtain $y = 1$. On most occasions, you may be left where the first equation involves both y and z . This would be similar to the first problem shown above.

- **Solve the remaining equations:** As $y = 1$ and $4y + 3z = 7$, we once again substitute $y = 1$ to obtain $3z = 3$ and so $z = 1$.
- **Once we solved for a variable, we trace back until we get all variables:** As $y = 1$ and $z = 1$, we substitute into one of the first few equations to get $x = 2$.

So, we obtain the solution $(x, y, z) = (2, 1, 1)$.

These problems are all about getting used to them. Try the following problems below until you think you got this! Solve the following with the substitution method.

Solve for (x, y) :

$$\begin{aligned} 10x + 12y &= -26, \text{ and} \\ -6x + 6y &= -24. \end{aligned}$$

Solve for (x, y, z) :

$$\begin{aligned} x - 2y + z &= -1, \\ x + 2y - z &= 7, \text{ and} \\ x + y + z &= 2. \end{aligned}$$

Solve for (x, y, z) :

$$\begin{aligned} -2x + 2y + z &= 14, \\ 3x - 2y + z &= -5, \text{ and} \\ -x + y - 2z &= -8. \end{aligned}$$

※ Elimination Method

While substitution works perfectly, sometimes the equations are set up in a way that makes substitution messy (like having to deal with lots of fractions). Elimination is a powerful shortcut where we add or

subtract the equations directly to make one of the variables “disappear.”

Think of this as a game of balance. If you have two true statements, and you perform the same operation to both sides, the statement remains true. By adding or subtracting the equations, we can cancel out a variable entirely.

The goal is to get the same number (or opposite numbers) in front of one of the variables.

- Line them up: Write the equations so that x , y , and z are in the same columns.
- Match them up: Multiply one or both equations by a constant so that the coefficients of one variable are opposites (e.g., $3x$ and $-3x$).
- Add or Subtract: Combine the equations to eliminate that variable.
- Solve and Back-Substitute: Once you have a smaller equation, solve it and plug that value back in to find the rest.

For example,

$$\begin{aligned} 2x + 3y &= 12, \text{ and} \\ x - 3y &= 3 \end{aligned}$$

Notice that the y terms are $3y$ and $-3y$. They are perfect opposites! Add the two equations together:

$$(2x + x) + (3y - 3y) = 12 + 3 \implies 3x = 15 \implies x = 5$$

Now that we know $x = 5$, substitute it into the second equation:

$$5 - 3y = 3 \implies y = \frac{2}{3}.$$

So, the solution is $\left(5, \frac{2}{3}\right)$.

Now, that we covered elimination, go back and solve the previous problems with the elimination method rather than substitution!

✂ In Context

Many problems won't explicitly state you are dealing with equations. On many occasions, you will have to create the equations yourselves. This is pretty similar to the variables handout where you have to try and figure out what to assign a variable for. Usually, try setting variables for unknowns and generally things you want to figure out. Let's try and look at a good example.

In a stationery store, pencils have one price and pens have another price. Two pencils and three pens cost 78 ¢. But three pencils and two pens cost 72 ¢. How much does one pencil cost?

Here, I want to solve for the price of a pencil. I would set a variable for that. But, I also don't know the price of a pen. So, let us set a variable for that too.

Denote the price of a pen and pencil as a and b , respectively. As two pencils and three pens are 78 ¢ then

$$2b + 3a = 78.$$

Likewise, three pencils and two pens are 72 ¢ so

$$3b + 2a = 72.$$

We have two equations and two unknowns which is perfect. If we had two equations but more than two unknowns then we wouldn't be able to solve it! This is the general rule of thumb: if we have more unknowns than equations, we won't be able to solve it. At the very least, not every variable. Some variables, we may still be able to solve for.

Back to the problem, we will use elimination but feel free to try substitution.

$$2b + 3a = 78 \iff 4b + 6a = 156,$$

$$3b + 2a = 72 \iff 9b + 6a = 216.$$

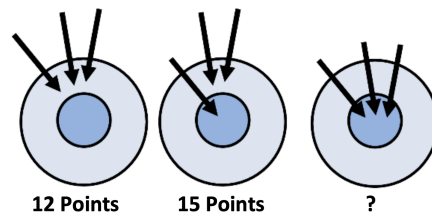
This comes from multiplying the equations by 2 and 3, respectively. Nicely enough, we can eliminate the a variable to get:

$$5b = (9b + 6a) - (4b + 6a) = 216 - 156 = 60 \implies b = 12.$$

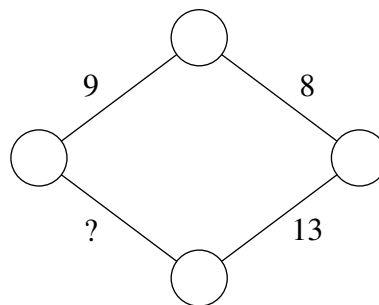
So, a pencil costs 12 ¢.

※ Practice Problems

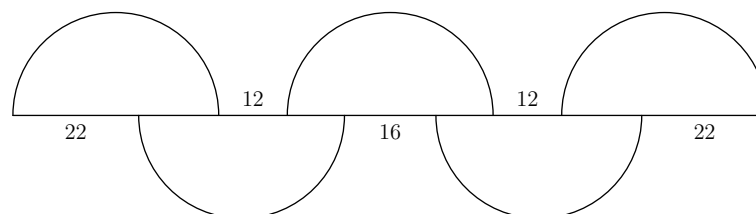
1. If 20 is added to one-third of a number, the result is the double of the number. What is the number?
2. A woman spent two-thirds of her money. She lost two-thirds of the remainder and then had \$4 left. With how much money did she start?
3. In a math contest of 10 problems, 5 points was given for each correct answer and 2 points was deducted for each incorrect answer. If Nancy answered all 10 problems and scored 29 points, how many correct answers did she have?
4. Six dollars were exchanged for nickels and dimes. The number of nickels was the same as the number of dimes. How many nickels were there in the change?
5. Diana shoots 3 darts, three times at a target board with two fields. The first time she scores 12 points, the second time 15. The number of points depends on which field she has hit. How many points does she score the third time?



6. Werner wants to label each side and each corner point of the rhombus shown with exactly one number. He wants the number on each side to be equal to the sum of the numbers on the corner points of that sides. Which number is he going to write in the place of the question mark?



7. Asha is three times older than Beth and Beth is two times older than Chloe. If Asha is 24 years older than Beth, how many years older is Beth than Chloe?
8. The diagram shows 5 equally big semicircles and the length of 5 distances. What is the radius of one semicircle?



9. In the addition problem below, A , B , and C are digits. If C is placed in the tens column instead of the units column as shown at the far right, the sum is 97. What are the values of A , B , and C ?

$$\begin{array}{r} A \quad B \\ + \quad C \\ \hline 5 \quad 2 \end{array} \qquad \begin{array}{r} A \quad B \\ + \quad \quad C \\ \hline 9 \quad 7 \end{array}$$

10. Suppose K , L , and M represent the number of points assigned to the three target regions shown at the right. The sum of K and L is 11, the sum of L and M is 19, and the sum of K and M is 16. How many points are assigned to M ?

